

1. Microscopy Image Processing & Blood Cell Quantification

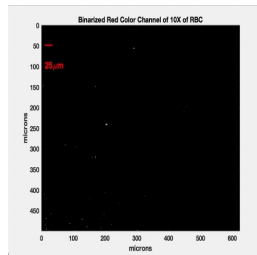
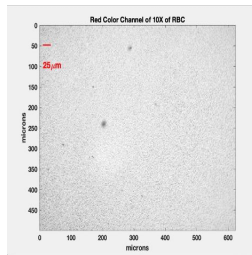
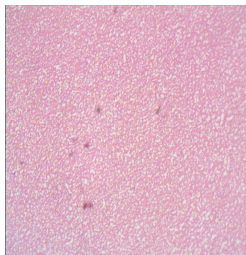


Image Processing

RGB analysis, grayscale conversion, thresholding, and filtering



Cell Segmentation & Detection

Segmentation, particle analysis, detection



Quantification

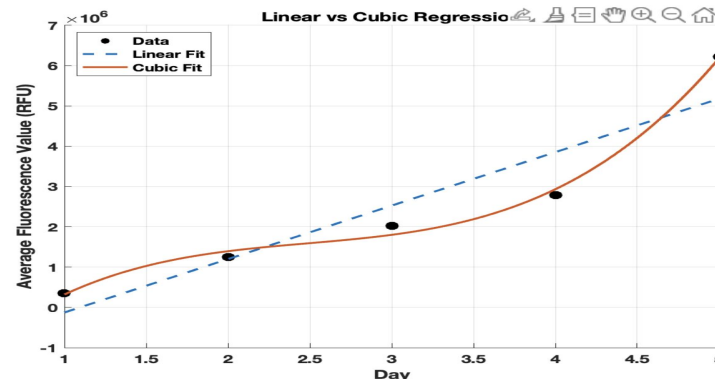
Automated cell counting and size measurements



Fourier Analysis

Frequency domain analysis to evaluate image patterns

2. Fluorescence Growth Analysis & Regression Modeling



Statistical Analysis

ANOVA testing and post-hoc comparisons



Data Summary

Average fluorescence with SEM error bars



Regression Modeling

Linear and cubic regression to model growth trends



Model Comparison

R^2 evaluation and model performance analysis